LYLE OKOTH

DATA ENGINEER - ML DATA PLATFORMS AND SPEECH CORPORA

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PROFESSIONAL SUMMARY

Data and Software Engineer with **6+ years** building the pipelines and platforms that make speech data trainable: ingestion, cleaning, ASR-assisted pre-labelling, reviewer workflows, versioned releases and the APIs that serve them.

Automated ingestion of **200 hours of audio per week at 98% label agreement**, built a platform hosting **18 African-language datasets and 40,000 labeled samples**, and delivered a **120-hour** aligned training corpus.

TECHNICAL SKILLS

- **Pipelines and orchestration:** Apache Airflow, Celery, Python ETL, batch and streaming jobs, retry-safe stages, incremental loads, de-duplication by content hash, dataset versioning
- **Storage and data modeling:** PostgreSQL, SQL data modeling, DuckDB, Parquet, Pandas, Polars, Elasticsearch, Redis, S3, Cloudflare R2, MinIO, Azure Blob
- **Data quality and annotation ops:** Label Studio, task schemas, reviewer rounds, inter-annotator agreement, gold sets, per-annotator accuracy, consent and licence records, dataset cards
- **ML data preparation:** FFmpeg, resampling, silence and noise filtering, loudness normalisation, forced alignment, diarization segmentation, train/val/test splits, COCO / LJSpeech / CVAT exports
- **Platform engineering:** FastAPI, domain-driven design, REST and SSE, Docker, GitHub Actions, AWS, Python, SQL, React

WORK EXPERIENCE

Machine Learning Researcher | SautiFlow Labs — 2026 - Present

Diffusion and flow-matching speech synthesis for languages with almost no public data.

- Trained a flow-matching TTS model covering **8 Kenyan languages** at **0.2s real-time factor** on a single consumer GPU, bringing multilingual synthesis inside interactive latency budgets.
- Cut **character error rate from 21% to 9%** on in-house Swahili and Sheng evaluation sets through fine-tuning, corpus curation and accent-aware text normalisation.
- Built a **120-hour** transcribed and force-aligned speech corpus from raw field recordings - the training base for the lab's synthesis and recognition models.

AI Product Developer | Freelance — 2026 - Present

End-to-end ownership on a small number of products: research, flow design, model, launch, pricing, operation.

- Built **Voxa** alone from brief to live product: speech-skit generation, account system, **3 billing paths** and a studio interface for creators.
- Made infrastructure portable by design - storage swappable across local disk, **R2, S3 and Azure** with **no application code changes**.

Machine Learning Engineer | SautiFlow Labs — 2022 - 2026

Owned the path from checkpoint to service: pipelines, inference APIs, evaluation harnesses, cost control.

- Shipped the SemaScribe speaker-diarization pipeline, processing **50,000 hours of meeting audio at 12% diarization error rate**.
- Served speech inference on FastAPI + Celery + Redis at **8,000 requests/day at 180ms p95**, using ONNX Runtime, dynamic batching and GPU worker pools.
- Cut **training spend by 40%** with mixed precision and preemptible GPU capacity at **equal model quality**, verified on frozen evaluation sets.

- Automated ingestion of **200 hours of audio per week** with **98% label agreement across 3 reviewers** via ASR pre-labelling and reviewer rounds.

Software and Data Engineer | SautiFlow Labs **2022 - 2025**

Two sides of one job: move data reliably, and ship a product people can actually use.

- Built SnapSavanna, a dataset and labeling platform hosting **18 African-language datasets and 40,000 labeled samples** with content-hash versioning.
- Designed domain-driven FastAPI services split into **5 bounded contexts** with ports and adapters at the edges, keeping model code out of the product path.
- Shipped SuraYa and EchoBuni with Google OAuth, **M-Pesa and card billing**, and swappable object storage across **R2, S3 and Azure**.
- Delivered server-sent events streaming for progressive audio generation with **first audio chunk under 400ms**.
- Built Airflow orchestration over Pandas, Parquet and DuckDB for ingestion, de-duplication and split generation.

SELECTED PROJECTS

LabelSet - Data and Labeling Platform

African-first corpora with ASR and detector pre-labelling, reviewer rounds, inter-annotator agreement, consent records and COCO / LJSpeech / CVAT export layouts.

Tech: Python, FastAPI, PostgreSQL, DuckDB, Apache Airflow, MinIO, Label Studio, Whisper, YOLO, React

EchoiID - Speaker Diarization and ASR

Who-spoke-when plus per-speaker transcription on stored files or live audio: voiceprint embeddings, clustering with overlap handling, word timestamps, SRT/VTT/JSON export.

Tech: PyTorch, pyannote.audio, Whisper, FastAPI, WebSockets, Redis, Celery, PostgreSQL, Docker, AWS

EDUCATION

BSc Computer Security and Forensics | Jaramogi Oginga Odinga University of Science and Technology (JOOUST) **[20XX - 20XX]**

ML, signal processing and security coursework; final-year project on audio watermarking for corpus provenance.

Mechatronics Engineering | Jomo Kenyatta University of Agriculture and Technology (JKUAT) **[20XX - 20XX]**

Fourier analysis, sampling theory, filter design and real-time embedded constraints - the grounding behind latency work.

OPEN SOURCE, WRITING AND COMMUNITY

- **Open ML releases:** 4 models and 3 datasets on Hugging Face, **12,000+ downloads**; model cards and training recipes per release.
- **Writing and tooling:** speech AI articles on Medium (@lyle-okoth); Python libraries for the YouTube and Google Drive APIs on PyPI.
- **Community:** OMDENA volunteer (2024); co-founder, SautiFlow Labs (2022).